Manish Jha

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LinkedIn | GitHub | Portfolio

Professional Summary

B.Tech CS graduate specializing in Generative AI, with hands-on experience building scalable LLM applications using Python and LangChain. Adept at data preprocessing and deploying AI-driven agents to solve complex business requirements.

Projects

CredIntel – AI-Powered Credit Analysis GitHub | Live Demo
Python, Pandas, PDF/OCR Processing, Groq API, Streamlit, Plotly

- **Data Processing**: Built a pipeline to extract financial information from PDF statements using pdfplumber, OCR, and LLM-assisted processing.
- **Financial Analysis**: Processed extracted data to calculate financial indicators and flag potential inconsistencies for credit analysis.
- **Dashboard**: Deployed a Streamlit dashboard with Plotly visualizations for financial data exploration and analysis.

DemandIQ – Retail Demand Forecasting GitHub | Live Demo
Python, Pandas, Scikit-learn, XGBoost, Flask, Gunicorn

- **Forecasting**: Developed a sales forecasting pipeline using 3M+ records across 54 stores and 33 product families, incorporating calendar and external features.
- **Evaluation & Deployment**: Benchmarked regression models and deployed XGBoost with preprocessing, reporting 94.99% R^2 on 26,730 test points.

HealthGuardAI – Healthcare Claims Validation GitHub
Technocrats Hackathon | Duo Project | AI/ML Contribution

- **Anomaly Detection**: Contributed to the AI/ML component to identify unusual claim amounts and duplicate billing, reporting 91.4% precision.
- **Claims Validation**: Worked on ML-based validation workflows to flag potentially abnormal healthcare billing patterns for review.

Chronic Kidney Disease Prediction GitHub
SmartBridge Experiential Learning – AI & ML

- **Model Development**: Preprocessed clinical data and compared Logistic Regression, Random Forest, and SVM; Logistic Regression achieved a reported recall of 1.0. Also explored ANN classification.
- **Deployment**: Integrated the selected model into a Flask application using Pickle, with an HTML/CSS interface for patient inputs and predictions.

GenAI Financial Assistant GitHub
Python, Django REST Framework, React, LangChain, FAISS

- **Financial AI**: Building a financial assistant that processes expense data and uses RAG to answer natural-language queries about transaction history.

Customer Retail Analysis GitHub
Python, Pandas, Scikit-learn

- **Classification**: Analyzed 541K+ e-commerce records and benchmarked Logistic Regression, Decision Tree, and KNN, reporting 91.3% accuracy on a 20,000-record holdout set.
- **Model Evaluation**: Used confusion matrices to investigate class imbalance and identify Macro-F1 and training-only resampling as evaluation improvements.

Technical Skills

Languages: Python, SQL, C++, C, JavaScript
Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Feature Engineering, Model Evaluation, Anomaly Detection
AI & Data: NLP, LLM Applications, RAG, OCR, Data Preprocessing, Pandas, NumPy
Generative AI: Vector Embeddings, Pydantic, Llama3 (via groq)
Frameworks: Flask, Django REST Framework, FastAPI, Streamlit
Tools & Databases: Git, GitHub, PostgreSQL, REST APIs, Plotly, Docker, Render, Streamlit

Education

D Y Patil University Ambi, Pune
B.Tech in Computer Science Engineering (AI & ML) 2022–2026
CGPA: 8.95

Achievements

Google Cloud: Selected for Google Cloud Innovator Program – Cloud & AI Track.
Ideathon: Top 10 finalist among 100+ teams.
ISRO IIRS: Certified in Remote Sensing & Image Analysis.